



**8th INTERNATIONAL
(FREE MISSION) UNMANNED AIR VEHICLES COMPETITION
PROJECT PRESENTATION REPORT**



TEAM NAME: Team Irtifaa	
VEHICLE TYPE: Fixed-Wing	
VEHICLE DEVELOPMENT FORMAT	Novel Vehicle
INSTITUTION NAME: National University of Sciences and Technology	
TEAM RESPONSIBLE: Ahmed Raza	

PROJECT TITLE: Development of a Fixed Wing UAV used for border surveillance, equipped with Object Avoidance Technology.



1. TEAM INFORMATION:

Faculty Advisor			
Full Name	Research Areas	Research Funding	Experience
Dr. Kashif Javed	• Agriculture • Healthcare • UAVs	• Autonomous Indoor Environment Management System by NCAI (Rs. 14.5m).	• Supervision of Master and PHD students. • Participation in International Workshops.

Students			
Full Name	Expertise	Duties	Experience
Ahmed Raza (Team Lead)	• Software, such as XFLR5, MachUp, MotoCalc, and SolidWorks. • Aerodynamic analysis and general sizing of UAV.	• Communication link between team and university. • Overseeing financial aspects. • Design and optimization of wings and tails.	• Designed wings and tails in IMechE National UAS Challenge and National Design, Build, and Fly (NDBFC). • Active member of Formula Student and BAJA SAE teams.
Muhammad Umer	• Software, such as MachUp, SolidWorks and Ansys.	• Creation of model using CAD. • Analysis of model using Ansys.	• Participated in IMechE National UAS Challenge. • 2nd position in ASME Design Challenge.
Muhammad Usama	• Software, such as SolidWorks and MachUp. • Working of dynamic systems.	• Design and optimization of fuselage. • Handling of manufacturing aspects of UAV. • Chart up testing plan for UAV.	• 2nd position in design phase in regional IMechE AKDC. • Successfully fabricated and tested Robotic Arm. • Manufactured payload mechanism of NDBFC.
Muhammad Hamza	• Interfacing python API's with hardware like depthAI, NAOqi API. • Altium PCB designing. • Deep learning. • Object detection. • Proteus.	• Bridging the gap between electrical and software teams. • Communicating between the Raspberry Pi and Pixhawk via MAV Link and drone kit API. • Implementation of CV on OAK-D.	• Successfully fabricated and tested obstacle avoiding car using CV. • Manufactured fully autonomous quadcopter.
Muzammil Ahmed	• Proficient in using circuit simulation in Proteus and PSpice. • Simulation of flight mechanics in MATLAB.	• Bridging the gap between electrical and mechanical teams. • Deciding the propulsion of UAV. • Calibration of control surfaces.	• Implemented autonomous and RC systems in IMechE National UAS Challenge and NDBFC. • Manufactured fully autonomous quadcopter.
Hasan Abbas	• Dataset collection and validation. • Development of custom neural network architecture for required needs. • Model testing.	• Work on architecture of the model and perform hyperparameter tuning. • Collect data for training, run data preprocessing and annotations. • Conduct tests for the predictions made by the model.	• Worked on crop harvest detection using normalized difference vegetation index. • Time series analysis using Facebook prophet.

Zain-ul-Abideen	• Big Data • Deep Learning • Computer Vision • GNNs • NLP • Forecasting models • Content relevance • Real-time ML systems	• Train a YOLOv7 object detection model to detect intruders. • Develop an algorithm to efficiently make predictions on image frames. • Calculate the GPS coordinates of intruders.	• Undertook 13 projects on Upwork and LinkedIn relating to CV. • Developed a motion detection algorithm by extracting 32 body points using Blazepose. • Built an Exercise recognition engine based on ConvLSTM.
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2. MISSION INFORMATION:

a. Mission Definition:

The mission of the UAV is to takeoff autonomously and then loiter in a circuit pattern over a specified area. In that area, the down pointing camera of the UAV will be used to monitor the ground area. It will be recording a footage of the area and analyzing the footage for any intruders. If an intruder is detected the UAV will send a SOS message and geo-coordinates of the intruder to GCS. During this flight the UAV will also be processing data from its front camera to detect any objects in its flight path. In case of any object being detected, it will change its path to avoid the object and then continue its regular route.

b. Mission Originality:

Our mission offers multiple unique features. These are listed below.

- Our UAV will avoid real obstacles whose location and dimensions are unknown to UAV. Therefore, through Computer Vision (CV) the UAV will navigate its way past the obstacles.
- The flight will be carried out in fully autonomous mode.
- There will be detection of persons (a friendly member and an intruder) on ground through CV.
- After detection, our UAV will also differentiate between the friendly member (security guard) and the intruder by comparing their GPS coordinates.
- GPS coordinates of the intruder will be sent to GCS.

These features aren't available in missions of previous competitions. A detailed comparative study is displayed in the table below.

Competitions	Mission	Dissimilarities
International UAV Competition (2022) held in Turkey	• Circuit Pattern. • Drop zone detection • Package Delivery	• Tasks can be performed semi-autonomously. • No Obstacle Avoidance. • Detection of drop zone only.
International Free Mission UAV Competition (2022) held in Turkey	• Fire quenching. • Dropping of life jackets. • Firing of bullets.	• Detection of targets only. • No Obstacle Avoidance.

Annual Student Unmanned Aerial Systems Competition (SUAS) held in USA	• Package Delivery • Obstacle Avoidance • Drop zone detection • Mapping	• Tasks can be performed semi-autonomously. • The obstacles are virtual, and their GPS coordinates are known to the UAV and the GCS. (Competitions, 2022) • Detection of personnel and drop zone only.
IMechE UAS Challenge held in UK	• Package Delivery	• Tasks can be performed semi-autonomously. • No Object detection. • No Obstacle Avoidance. (CHALLENGE DOCUMENT LIBRARY, 2021)
IMechE UAS Challenge held in Pakistan	• Spraying of Chemicals.	• Tasks can be performed semi-autonomously. • No Object detection. • No Obstacle Avoidance.

- **Note:-** Due to limited data available, we couldn't contrast our mission with more of "International Free Mission UAV Competition" missions.

c. The Way the Mission Is to Be Executed in the Field:

In the field we will set up 6 poles with a height of 10m each. The poles will be setup in a pattern shown in fig. To make the poles more recognizable we will hang a Panaflex banners on each pole of about 2mx1m. The distance between poles on each level will be 7m and the distance between each level of the triangle will be 7m. The UAV will follow a path that goes through the obstacles. It will follow its waypoints and whenever it detects any object it will change its trajectory to go around the object and then return to its original path. In this way the UAV will be tested for its object avoidance capabilities.

In the same mission flight, we will carry out the UAV's intruder detection capabilities. Two members from our team will be impersonating the friendly member and the intruder. Both of them will be located at a distance from each other. The UAV will fly above them, differentiate between them, and send the GPS coordinates of intruder to GCS.

- **Note:-** The team members will have no special clothing etc. However, the friendly member will have a "NRFL2401 wireless transmission module" explained in the hardware section.

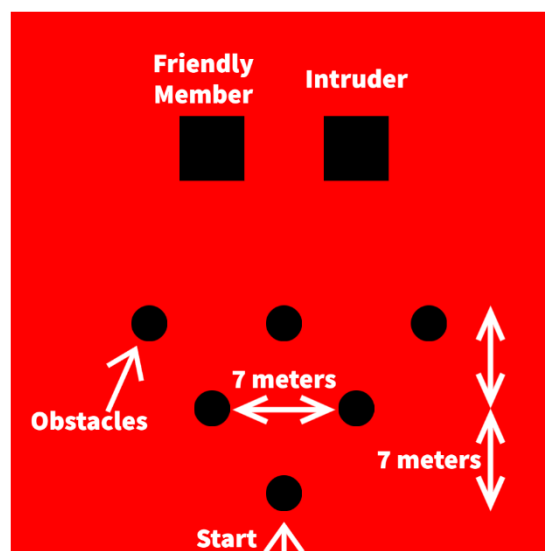


Figure 1: Mission Schematic

3. UAV INFORMATION:

a. Design:

The overall design for UAV has been intended to cater for three priorities: ease of manufacturing, performance, and cost -in that order. Therefore, the decisions are based on these priorities. Furthermore, the mission requirements dictated a UAV which can cruise at low speeds for good performance of the object avoidance and image detection model. Also, the UAV must be able to maneuver easily to avoid obstacles. The UAV must smoothly take-off and land. Lastly, the UAV must be modular for logistical purposes.

i. Wings and Tails

To that end, the overall estimated weight initially was, in addition to 10% contingency, 6.4kg. Several iterations were performed on a formulated general sizing spreadsheet encompassing various parameters (Hunsaker, 2022). The table below summarizes the results in the final iteration.

				Aircraft Dimensions and Specifications		Performance	
	Wing	Horizontal Stabilizer	Vertical Stabilizer				
Airfoil	MH 114	NACA 0012	NACA 0012	Length (m)	1.67	Cruise Speed (ms^{-1})	15
Span (m)	2.336	0.906	0.6	Width (m)	2.6	Max Speed (ms^{-1})	25.5
Area (m^2)	0.814	0.164	0.056	Height (m)	0.39	Stall Speed (ms^{-1})	8.95
Angle of Incidence ($^\circ$)	+0.7	-3.45	0	Weight (Kg)	6.0	Flight Time (min)	11
Mean Aerodynamic Chord (m)	0.349	-	-	Prop Size (in x in)	13x8	Min Turn Radius (m)	12
Aspect Ratio	6.7	5	6.43	Motor	OMA 5010/810 KV	Assembly Time (min)	5

A trade-off between high wing loading and low stall speed was made to fulfill the mission requirements. Furthermore, the aspect ratio (AR) is advised between 6-8 for these types of UAVs (Lennon, 1996). However, AR of 6.7 provided most efficient results in terms of modularity and drag of wings. The wing's sweep angle is kept 0° for ease of manufacturing. Moreover, several iterations were carried out for best taper ratio on XFLR5 and 0.4 extracted the optimum results. A wing's angle of incidence was set to provide enough lift to balance weight in cruise conditions. For the airfoil selection, we analyzed 4 different airfoils on XFLR5 on the Reynolds number (1,026,000) expected at our cruise speed: NACA-6412, MH-114, FX 63-137, and E-214 (Atmospheric Properties Calculator, 2016). Out of these, FX 63-137 and MH-114 provided the necessary lift coefficient to match our geometry in general sizing (Michael S. Selig, 1995). Furthermore, among these two airfoils, MH-114 provided the highest performance, as can be observed in figure, in terms of Cl/CD vs Angle of Attack (AOA) and C_m vs AOA.

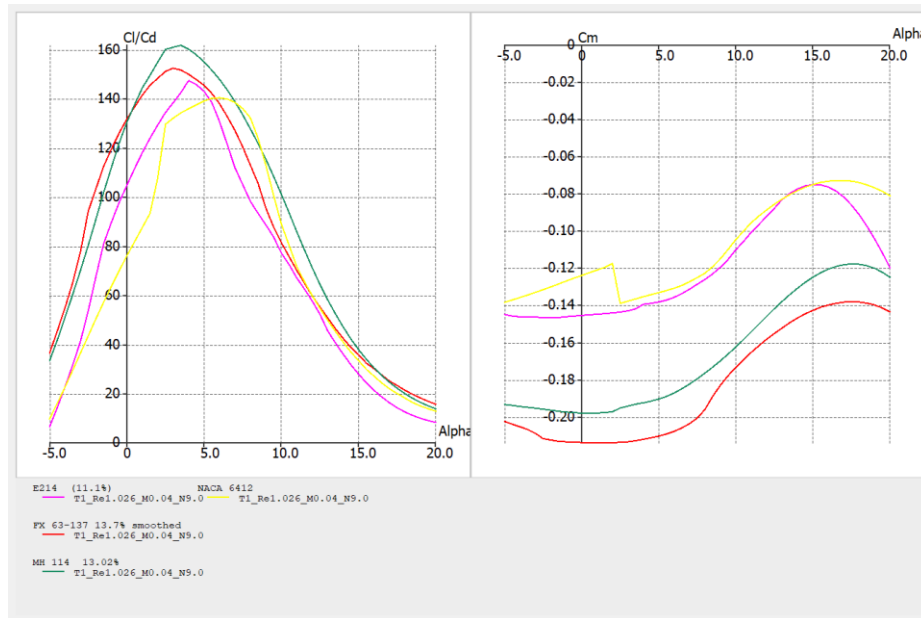


Figure 2: 2D Airfoil Comparison. MH-114 (Green), FX 63-137 (Red), NACA-6412 (Yellow), E-214 (Pink)

For the tails, it was advised in the course “Introduction to Fixed-Wing Drone Design” that we must use horizontal tail volume ratio between 0.3-0.6 (Hunsaker, 2022). We settled on a value of 0.55 through literature review (Lennon, 1996). Then, we iterated the design process to quantify the best combination of moment arm and area values that provided us with the finest blend of maneuverability, modularity, and stability. NACA 0012 was used for its symmetrical and predictable nature. We resolved for an inverted U-tail configuration since it provided a high tail efficiency of 0.9 (Lennon, 1996). The horizontal tail incidence angle was set considering airfoil pitching moment, wing’s drag moment, and thrust moment (Lennon, 1996). The same process was carried out for vertical tail.

For the manufacturing of wings and tails, we have opted to use Extruded Polystyrene Foam (XPS) due to its cost, strength, and weight. After placing airfoil pieces on each end, hot wire cutting of XPS will be done. Then, holes will be drilled into the wings to insert carbon fiber spars into it using epoxy and German glue. The two wings’ carbon fiber spars will be connected with each other using aluminum couplers and flat head dowel pins, as shown in figure 3. This will ensure modularity of aircraft. Lastly, the wings will be covered with fiber glass cloth layers since our mass fraction exceeds the limit ($37.27\% > 35\%$) (Aero lab, 2015). The vertical tail will be mounted above the 3D printed tail junction, not as shown in the figure 3.

ii. Fuselage

The fuselage is design to give a large amount of space to add components, which allows for the center of mass to be adjusted as needed. Additionally, the large space allows for easy access to internal components, allowing them to be swapped with minimal effort when necessary. It has the length of 110cm from the rear back to the tip of the nose and has an internal diameter of 19 cm. It is built on nearly square cross-section, which is easy to build. The nosecone is constructed in the same cross-section coupled up to the 16 cm. Main wing is mounted transversely on the rear of the fuselage with the help of carbon fiber spars and is mounted 10 cm forward from the very trailing edge of the fuselage, this is implemented for mounting landing gear. There are two cameras, as shown in figure 3, one facing the front for obstacle avoidance and the other facing the ground for intruder detection. Also, the fuselage has two hatches one at the front and the other at the back the front hatch could be also quickly secured or opened without any tools for easy access to component’s compartment, the rear hatch is built for assembling of the wings right above the wing mount place, which is internally accessible. The fuselage will be made of fiber glass cloth using a mold made out of XPS foam.

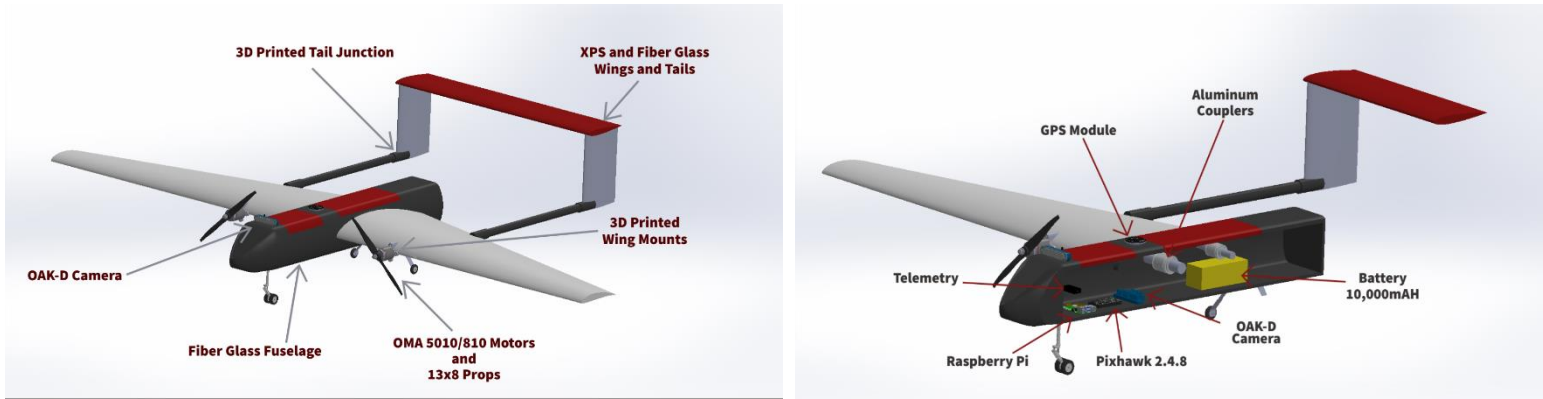


Figure 3: (Left) Overall Design of UAV. (Right) Internal Spacing and Items of UAV.

iii. Propulsion

Preliminary thrust and endurance specifications from the MATLAB sizing algorithm were further explored using the MotoCalc tool for motor and prop selection. The analysis was limited to the motors locally available so to avoid delay and cost associated with import of products. The final iterated UAV propulsion system consists of two OS Motors OMA 5010/810KV motors each paired with an APC 13x8E propeller. This combination provides 5.3 kg of static thrust using 106.8 amps of current at max throttle. At the best lift-to-drag ratio airspeed, the excess-thrust (34.26N) to weight (60.15N) ratio is 0.57:1, which will give steep climbs and excellent acceleration. This configuration has sufficient in-flight thrust for almost any aerobatic maneuver. The speed controllers are rated for 60A, providing more than enough allowance for current spikes. These parts, along with all the other electrical components in the plane, are powered by a two 4S 10,000 mAh 14.8V Li-Po batteries.

b. Hardware:

i. The Flight Controller:

The Pixhawk 2.4.8 Flight Controller will be used by the UAV. Autonomous takeoff, flying, and landing are among its characteristics. Additionally, it enables both automated and human PID value adjustment of the aircraft to enhance overall stability when autonomous flight. It also has a sturdy construction and dependability. The Pixhawk 2 has capabilities not available on the Pixhawk 1 and APM board, allowing the team to send and receive orders across various telemetry connections, such as both telemetry ports and the USB port, in order to interface with the aircraft.

The flight controller has an internal magnetometer, gyroscope, and accelerometer whose values are constantly being reported to the Ardupilot at the ground control system. This enables the UAV to fly steadily during its mission.



Figure 4: Pixhawk 2.4.8
(<https://ardupilot.org/copter/docs/common-pixhawk-overview.html>)

ii. Object Detection:

1. Raspberry Pi:

Since detecting an intruder requires a machine-learning approach, people in covered areas need to be detected precisely for further processing. The Pixhawk has a flash memory of 2 MB and 32-bit RAM and is therefore incapable of processing images and taking all the algorithmic decisions. For such a task, a companion computer becomes necessary. In this case, that computer is the Raspberry Pi 4B. This module has the capability of video decoding at up to 4Kp60, 8GB RAM, dual-band 2.4/5.0

GHz wireless LAN, Bluetooth 5.0, Gigabit Ethernet, etc. With these specifications, image processing is no longer a problem, and Pixhawk can become more versatile.

The flight controller can communicate with the Raspberry Pi through the MAVLink protocol over a serial connection. The pin assigned for this purpose is TELEM 2 which is a serial port of the Pixhawk. MAV Proxy can be used to send commands to the flight controller from the Pi. It can also be used to route telemetry to other network endpoints. To give proper instructions about the motion of the UAV according to the input values from the sensors, we have a proper Python API known as the drone kit.

By using the drone kit open-source API, we can take advantage of various built-in functions to control the UAV, providing us with control options not just from the controller, but also from a programming framework. In this way, the UAV also monitors the messages sent by the Raspberry Pi using Python.

2. OpenCV AI Kit: OAK—D

An external camera OAK-D is now connected to the Pi so that object detection can be performed using Python. OAK—D is a spatial AI powerhouse, capable of simultaneously running advanced neural networks while providing depth from two stereo cameras and color information from a single 4K camera in the center. The depth AI library is used to communicate between the OAK device and the python (Raspberry PI).

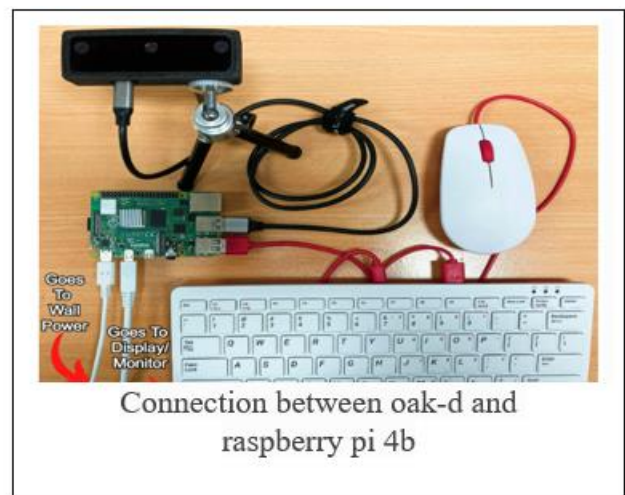
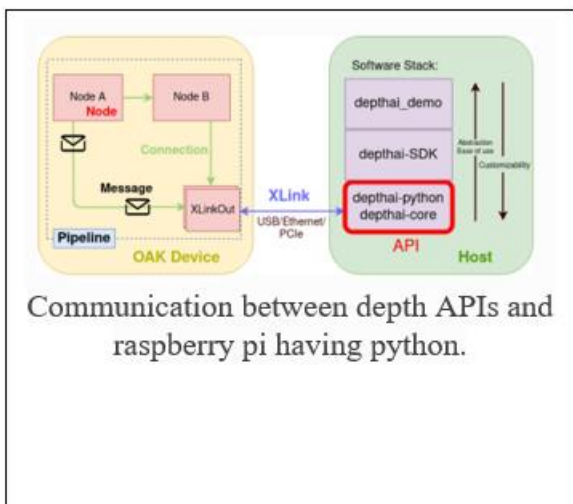


Figure 5:(Left) Communication between depth API and Raspberry Pi (<https://docs.luxonis.com/projects/api/en/latest/index.html>). ((Right) Raspberry Pi 4B and OAK D Connection (<https://core-electronics.com.au/guides/oak-d-lite-raspberry-pi/>).

Python on the raspberry pi is now fed a model to detect people and their coordinates, as explained in the **software section**. An intruder can then be detected using these coordinates.

iii. Intruder Detection:

Once a person is detected using the Raspberry Pi described in the first section, the next step is to determine whether that person is an intruder or not. The approach used for this is simple. A piece of hardware consisting of an Arduino Nano, 2.4GHz NRFL2401, and Neo 6M GPS module is given to non-intruders. The NRFL2401 wireless transmission modules are designed to send the location of the non-intruders to the companion computer (Raspberry Pi) on the drone where we have an NRFL2404 PA/LNA module to receive those coordinates. This way a master-slave configuration is generated where multiple friends will send their coordinates to the master on the raspberry pi which will be compared by the raspberry pi with the coordinates of the person detected using computer

vision techniques as explained in the **software section**. This comparison enables us to check if the detected person is a friend or an intruder.

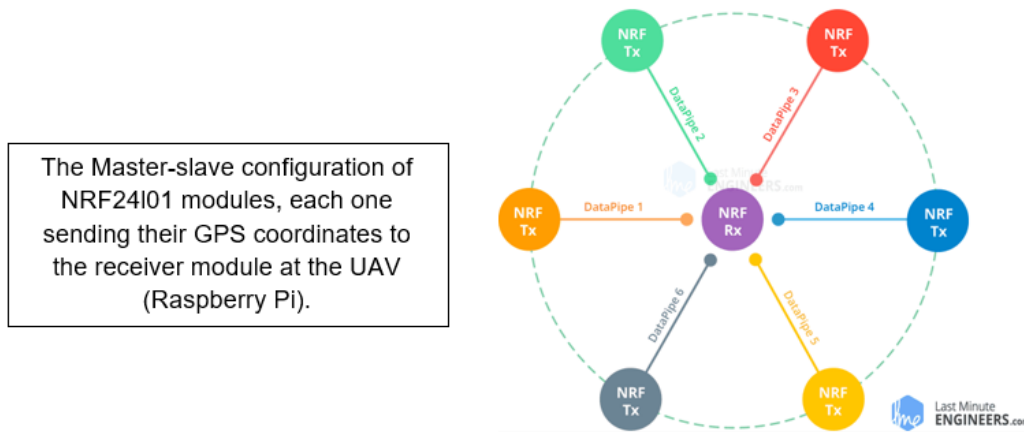


Figure 6: Connection of NRF24I01 Modules with Raspberry Pi 4B
(<https://lastminuteengineers.com/nrf24i01-arduino-wireless-communication/>)

iv. Obstacle Avoidance:

To avoid an obstacle, it is necessary to first detect an object and then find the distance from it. The UAV detects and measures distance with the help of another OAK-D camera which will be setup facing the forward direction. It will also be able to measure depth. To calculate this minimum distance, we use the depth formula.

$$\text{minimum distance} = \frac{\text{focal length in pixels} * \text{baseline}}{\text{disparity in pixels}}$$

Choose the maximum value for disparity in pixels parameter (it is inversely related, so maximum value will yield the smallest result).

For example, OAK-D has a baseline of **7.5cm**, focal length in pixels of **882.5 pixels** and the default maximum value for disparity in pixels is **95**, using the formula above gives the depth.

$$\text{distance} = \frac{882.5 * 7.5\text{cm}}{95} \approx 70\text{cm}$$

The OAK-D can measure up to 19 meters theoretically which is ideal for our task of avoidance. Now once the object is detected and the distance is calculated, the final step shows it working with the overall system.

The figure shows the OAK-D camera in depth mode. Here the x, y, and z coordinates show the coordinates of the person relative to the camera.

A z coordinate represents the actual distance between the camera and an object.

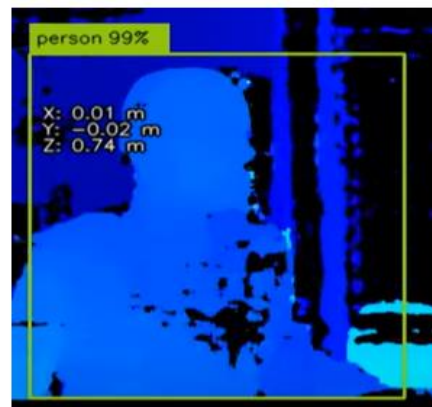


Figure 7: Depth Mode of OAK D
(https://www.youtube.com/watch?v=ZscWU6_vnQo)

v. Communication With Ground:

Now that all the components are attached to the raspberry pi who in turn is attached to the flight controller, the UAV is complete. But unfortunately, there is no communication between the ground system and the UAV. There must be a ground system to get continuous information about the autonomous mission and to know the coordinated and other states of the UAV.

To establish this wireless MAVLINK communication, we use a pair of 433MHz radio telemetry at both, the ground and UAV. At the ground, the telemetry communicates with mission planner software using the serial USB port. At the other end, the telemetry is attached to the TELEM 1 port of the flight controller. This enables serial communication between the telemetry and the flight controller.

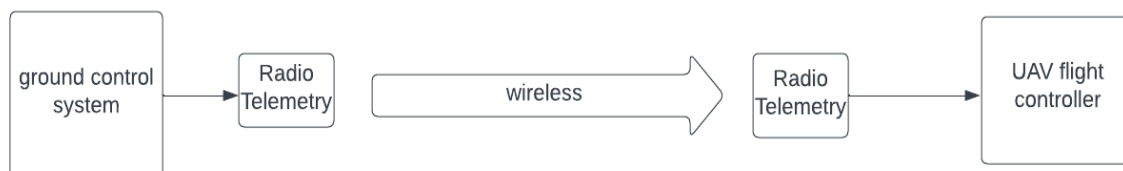


Figure 8: Schematic of Ground Communication

vi. The Big Picture:

Now all the hardware is linked together to make the mission work. A camera on the UAV will first recognize people and their coordinates using CV. At the same time, the friendly member will be sending their GPS coordinates to the Raspberry Pi connected to the UAV with the master module. This will compare these coordinates with those of the detected points using CV. This will enable us to detect an intruder along with its geo-coordinates which could be sent to the GCS as explained in **the software section**. Meanwhile, if any object is detected by the camera, steps are taken based on its position relative to the camera to avoid it. Figure 9 shows the complete workings of the system.

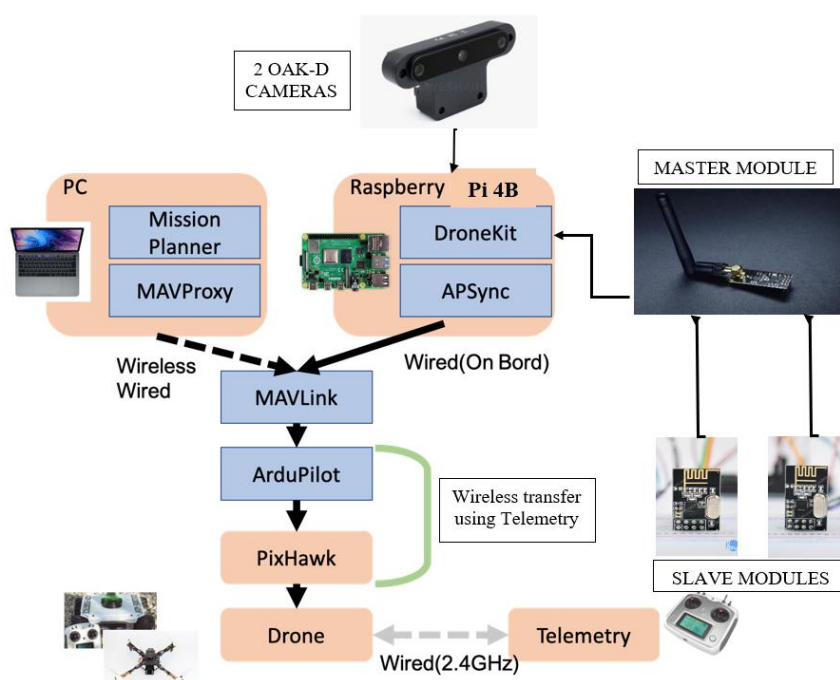


Figure 9: Total Communication Schematic

c. Software:-

i. Ardupilot: -

Ardupilot a flight control software will be used to control the UAV since it is not only opensource but also compatible with various devices. Particularly in Ardupilot, Arduplane firmware will be used due to the UAV type. It is important to properly calibrate the Pixhawk with the UAV for stabilize flight. Autotuning mode will be utilized to further the stability of the flight.

In addition to this, Pixhawk Mission Planner mode will be used for autonomous flight. Waypoints will be provided though the UBLOX NEO-6M V2 GPS module for the mission. We will interface the Raspberry Pi with the Ardupilot to change our waypoints mid-flight according to the object avoidance algorithm. The output from the camera will be used to decide the next waypoints and then Ardupilot will fly the drone according to the waypoints.

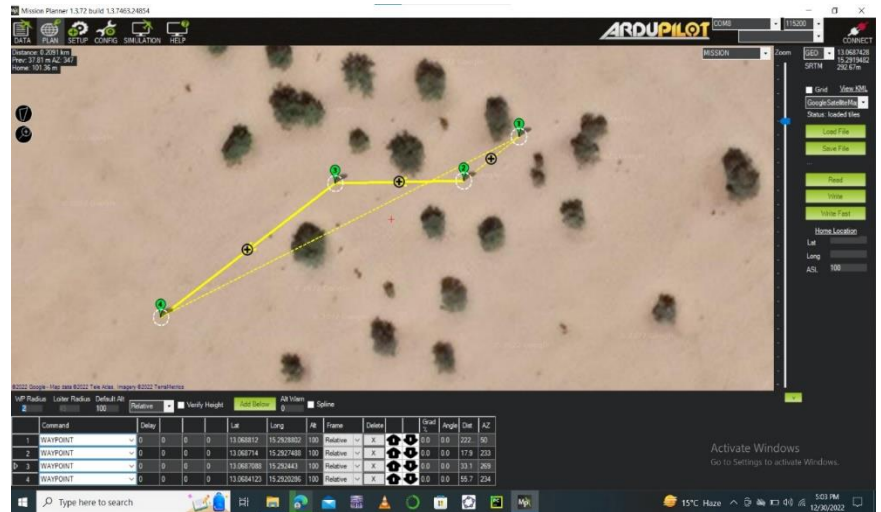


Figure 10: Waypoints on ArduPilot

ii. Computer Vision: -

YOLO (You Only Look Once) will be used to train CV models for both obstacle avoidance and intruder detection. It is a real-time object detection algorithm. It is designed to be fast and efficient, allowing it to process images and video in real-time and make object detection predictions.

Like other object detection algorithms, YOLO v7 divides the input image into a grid of cells, and each cell is responsible for predicting the presence and location of objects within its area. YOLO v7 uses anchor boxes to predict the size and shape of objects, and it uses a multi-scale prediction strategy to detect objects of different sizes in the image.

YOLO v7 is trained on a large dataset of annotated images and uses a variant of the Darknet architecture, which is a lightweight and fast CNN architecture. It is designed to be able to run on a variety of devices making it a top choice for real-time object detection in our mission.

1. Data Set Collection

We will be using two datasets VisDrone and Okutama-Action for fine-tuning our Yolov7 model. Fine-tuning is a process which involves training the model on a new dataset, using the pre-trained model as a starting point, and adjusting the model's weights and hyperparameters to optimize its performance on the new task. We will only be using the specific classes of these datasets related to humans in the frame. Furthermore, OpenCV and PIL library will be utilized to deal with image pre-processing. Along with the online dataset, a small percentage of the dataset will be formulated by our team. This will be done by employing our existing UAVs. The images collected will be annotated using Roboflow.

2. Testing

For testing our model, we will use Intersection over Union (IOU) and Mean Average Precision(mAP) as a metric. The IoU is defined as the ratio of the intersection of the detected bounding box and the Ground Truth (GT) bounding box and their union. Average Precision (AP) is calculated as the weighted mean of precisions at each threshold; the weight is the increase in recall from the prior threshold. Mean Average Precision is the average of AP of each class.

3. Intruder Alert using Triangulation

To calculate the GPS coordinates of a person using a camera, we will need to use the location of the camera and the distance to the person. This can be done using triangulation, which is a method of finding the distance to an object using trigonometry.

First, we will need to know the latitude and longitude of the camera, which will be obtained through Pixhawk 2.4.8. We will also require the knowledge of the altitude of the camera, which can be obtained from a digital elevation model.

Next, we will require the distance between the camera and the personnel which will be provided by the OAK-D camera.

Using this information, we can calculate the GPS coordinates of the personnel. NumPy arrays will be employed while computing the results regarding the GPS coordinates of the detected person. If the GPS coordinates of the friendly member found using the NRFL2401 match any of the GPS coordinates calculated by our camera to a specific extent, then we can say that the detected person is not an intruder. If the GPS coordinates of the known people found using the NRFL2401 do not match any of the GPS coordinates calculated by our camera, then we can say that the detected person is an intruder. After detection of the intruder, we will send the GPS coordinates of that intruder to the GCS to alert it. Hence further investigations can be done to deal with this discrepancy.

d. Indigenization: -

This year Team Irtifaa will participate in the competition for the first time. Thus, in order to ensure achievement of our mission, we have decided to use established techniques so as to reduce complexity as a tradeoff for effective implementation. This doesn't imply that we haven't been original. Some of our development has been described below.

i. GPS Coordinates: -

It is a common practice to use GPS modules to find the coordinates of objects. However, we are developing Computer Vision techniques to carry out this task, as explained earlier.

ii. Intruder Differentiation: -

The approach the team is utilizing to discern between the family member and the intruder is novel. The use of NRFL2404 PA/LNA module for continuous transmission of signals to UAV and the comparison between the coordinates of the family member and intruder are methods explored by the software team.

e. Innovation: -

We have invested much time to discover methods that separates our mission with that of others. To that end, the team has found some success.

i. Use of Depth Camera for distance: -

While undertaking literature review, we found that most teams employ LIDAR for distance calculation between the camera and an object. However, we resolved to use a depth camera for this cause. This was done for numerous reasons. Firstly, for measurement of distance for far away objects and faster processing. Secondly, Lidar only throws straight rays; therefore, it doesn't cover a wide area, which can be problematic for obstacle avoidance. Lastly, the camera provides the orientation of the object, which helps in the decision making of the trajectory of the UAV after an object has been detected in its path.

4. BUDGET DATA:

a. Budget Table: -

Components	Quantity	Cost (TL)	Total Cost (TL)	Reason for Purchase
ESC 60A	2	396.7	793.4	Throttle Control
OAK D Camera	2	4550	9100	Object Detection and Image processing
Battery (10,000 mAh 4s)	2	2025	4050	Supply power to electronic components
Servo Linkage	4	3	13	Connect pushrod to horns
Push Rods	4	2	10	To transfer servo torque to horns
Horns	4	5	18	To transfer torque to control surfaces
Metal Gear servo motors	4	45	182	To generate torque
Motors (OMA 5010/810kV)	2	1116	2231	To provide propulsion
Propeller (13x8)	2	95	190	To generate thrust
XT60 Connectors	4	17	66	Connection of battery to power module and ESC
Miscellaneous wiring	1	34	34	For connections of electronic components
Carbon Fiber Spar	6	620	3719	To connect tail to wings and wings with each other
Fiber Glass Cloth	2	869	1738	Layering of wings and fuselage over XPS foam
Insulation Foam Board (6'x3')	3	183	549	To manufacture wings and tails
Servo Extension	4	13	53	To connect servo with receiver resided in fuselage
Main and Nose Landing Gear wheels	3	69	69	To provide friction for taxiing
Main and Nose Landing Gear	1	736	736	To connect wheels with fuselage
Nails and Bolts	1	41	41	For fixtures
Battery Strap	1	41	41	To hold battery in its position
Transmitter/Receiver	1	1365	1365	For communication with UAV in case of emergency
Pixhawk	1	2358	2358	For autonomous flight
GPS Module	1	397	397	To provide GPS coordinates
Telemetry	1	1696	1696	To provide flight vitals
Shock Absorber Mount	1	66	66	For safety to electronic equipment in case of crash
Double tape	1	10	10	To hold electronic equipment in fuselage
Epoxy/glue	1	83	83	For fixtures and layering of fiber glass

Raspberry pi 4B	1	4137	4137	As a companion computer
Manufacturing Equipment	1	826	826	3D printing material, hot wire cutting equipment etc.
NRF Module	1	15	15	For transmission of signals from friendly members
Arduino Nano	1	103	103	To provide hardware for NRF module

b. Budget Parts:

Budget Item	Amount (TL)
Grand Total	34,667
Contingency Total (CUD)	38,134
Amount Spent (PSA)	11,088
Amount to be Raised by Team (OTA)	21,266
Requesting from TUBITAK (TRA)	5,780
TRA/CUD Ratio	0.152

- Note:- Contingency total is grand total with 10% addition in case of cost over-run.

Our team has also actively started approaching potential commercial sponsors. To that end, we have arranged meetings with some close companies, made a list of 250 potential sponsors, and recruited members for External Relations and Sponsorships for advice related to financing. Furthermore, to start the manufacturing of the UAV, the team has decided to pitch in seed money from their own means.

5) References: -

- Aerolab, U. (2015, October 7). Mission-Specific Aircraft Design (Part 1). Utah, United States of America.
- Atmospheric Properties Calculator*. (2016, January). aerospaceweb:
<http://www.aerospaceweb.org/design/scripts/atmosphere/> adresinden alındı
- CHALLENGE DOCUMENT LIBRARY*. (2021, September 16). imeche.org:
<https://www.imeche.org/docs/default-source/1-oscar/challenges/uas-challenge/uas-2022/uas-challenge-2022---competition-rules---issue-1-final.pdf> adresinden alındı
- Competitions*. (2022). suas-competition: https://suas-competition.org/s/auvsi_suas-2022-rules.pdf adresinden alındı
- Hunsaker, P. D. (2022, August). Introduction to Fixed-Wing Drone Design. Utah, United States of America.
- Lennon, A. (1996). *Basics of R/C Model Aircraft Design: Practical Techniques for Building Better Models*. Motorbooks Intl.
- Michael S. Selig, J. J. (1995). *Summary of Low-Speed Airfoil Data*. Virginia: SOARTECH PUBLICATIONS.